STAT-5080 HW2

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February 5, 2026

Problem 1

A **Completely Randomized Design** can be used to design the experiment.

Design Steps:

Step 1: Select sample sizes n_1, n_2, n_3, n_4 with $n_1 + n_2 + n_3 + n_4 = 100$

Step 2: Choose n_1 units at random to receive Drug A, n_2 units at random from the remaining units to receive Drug B, and so on.

Step 3: Randomize the order of experimentation for the 100 experimental units.

Our Design:

	Drug A (1)	Drug B (2)	Drug C (3)	Drug D (4)
1	1 (76)	3 (24)	2 (39)	6 (86)
2	4 (53)	5 (92)	10 (35)	8 (83)
3	13 (29)	7 (40)	11 (43)	9 (90)
4	16 (70)	19 (61)	14 (78)	12 (1)
5	29 (50)	22 (71)	15 (22)	18 (37)
6	35 (16)	23 (51)	17 (28)	20 (46)
7	39 (77)	24 (44)	25 (49)	21 (67)
8	40 (99)	27 (56)	31 (20)	26 (60)
9	43 (13)	28 (100)	34 (97)	32 (98)
10	45 (34)	30 (7)	38 (89)	36 (2)
11	46 (65)	33 (87)	50 (11)	37 (48)
12	53 (73)	41 (41)	52 (91)	42 (58)
13	58 (66)	44 (26)	54 (81)	47 (19)
14	59 (94)	55 (54)	57 (68)	48 (17)
15	61 (84)	56 (45)	62 (69)	49 (74)
16	71 (14)	60 (9)	67 (72)	51 (31)
17	72 (42)	63 (10)	68 (88)	65 (3)
18	78 (63)	64 (15)	70 (30)	69 (12)
19	80 (75)	66 (32)	73 (6)	74 (85)
20	84 (21)	76 (93)	77 (55)	75 (96)
21	86 (64)	85 (62)	81 (33)	79 (36)
22	89 (4)	88 (8)	82 (80)	83 (5)
23	94 (57)	92 (52)	90 (23)	87 (18)
24	98 (27)	93 (59)	91 (47)	97 (82)
25	100 (79)	95 (38)	96 (25)	99 (95)

Random assignment ensures that any systematic differences observed among groups can be attributed to the treatments rather than to pre-existing characteristics. Equal replication (n = 25 per drug) yields comparable precision across all treatment comparisons. In addition, randomization mitigates the risk of confounding treatment effects with time-related or operational factors, such as subject fatigue or equipment drift.

Problem 2

Measuring all samples of each soil type together seems more convenient and organized. However, **random ordering is essential for valid statistical inference**, and here's why:

1. **Eliminating "Nuisance Variables"**

   If all samples of Soil A are measured first and Soil D last, any external changes over time—such as **equipment drift** (the pH meter losing calibration), **temperature fluctuations**, or **technician fatigue**—will be "confounded" with the soil type. Randomization ensures these environmental factors affect all soil types equally rather than biasing a specific group.

2. **Ensuring Statistical Independence**

   Statistical tests (like ANOVA) require that each measurement be an **independent observation**. Measuring in blocks (A-A-A) creates a correlation between consecutive samples. If the first measurement of Soil A is slightly off, the next two are likely to be off in the same direction, which artificially reduces the estimated "experimental error" and makes the results scientifically invalid.

3. **Reducing Observer Bias**

   If the technician knows they are measuring the same soil three times in a row, they may subconsciously try to make the readings consistent. Randomizing the order "blinds" the technician to the expected result, ensuring the data reflects the true acidity of the soil rather than a desire for consistency.

Problem 3: Circuit Response Time Analysis

Table 1: Summary Statistics by Circuit Type

Circuit	n	Mean	SD	Variance	Min	Max
1	5	12.6	3.97	15.8	8	18
2	5	17.4	4.62	21.3	12	23
3	5	8.4	4.39	19.3	5	16

Part (a)

One-Way ANOVA Effects Model

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}$$

where:

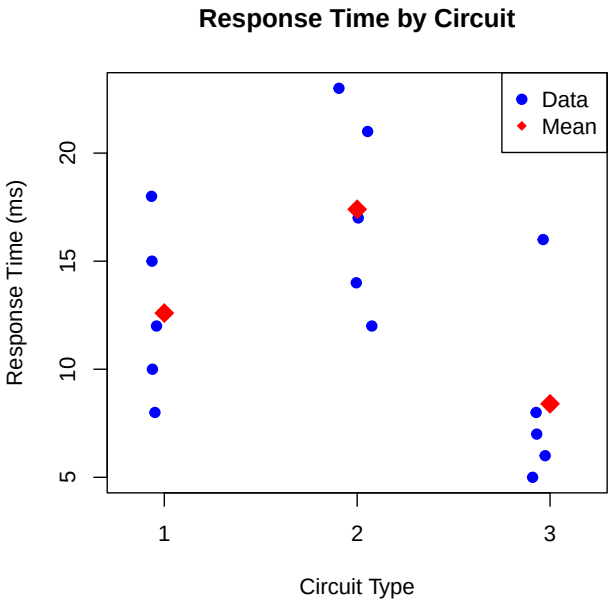
- Y_{ij} = response time for the j-th observation of circuit type i
- μ = overall mean response time
- τ_i = effect of circuit type i (treatment effect)
- $\epsilon_{ij} \sim N(0, \sigma^2)$ independently (i.i.d. normal errors)
- $i = 1, 2, 3$ (circuit types)
- $j = 1, 2, 3, 4, 5$ (replications)

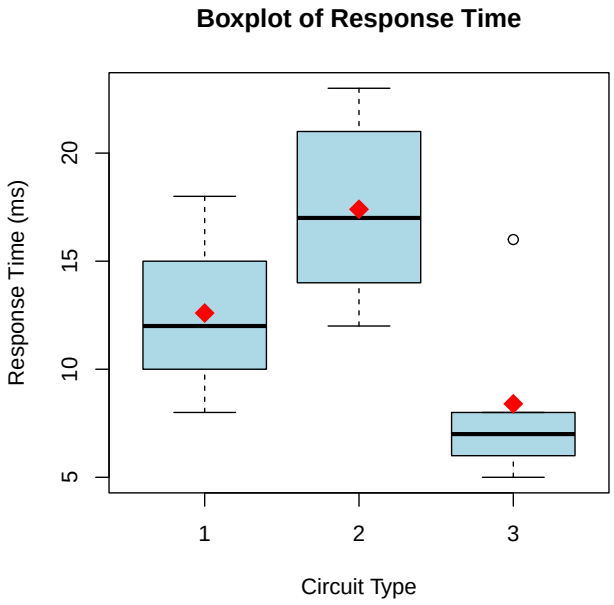
Constraint: Under the sum-to-zero constraint: $\sum_{i=1}^{3} n_i \tau_i = 0$

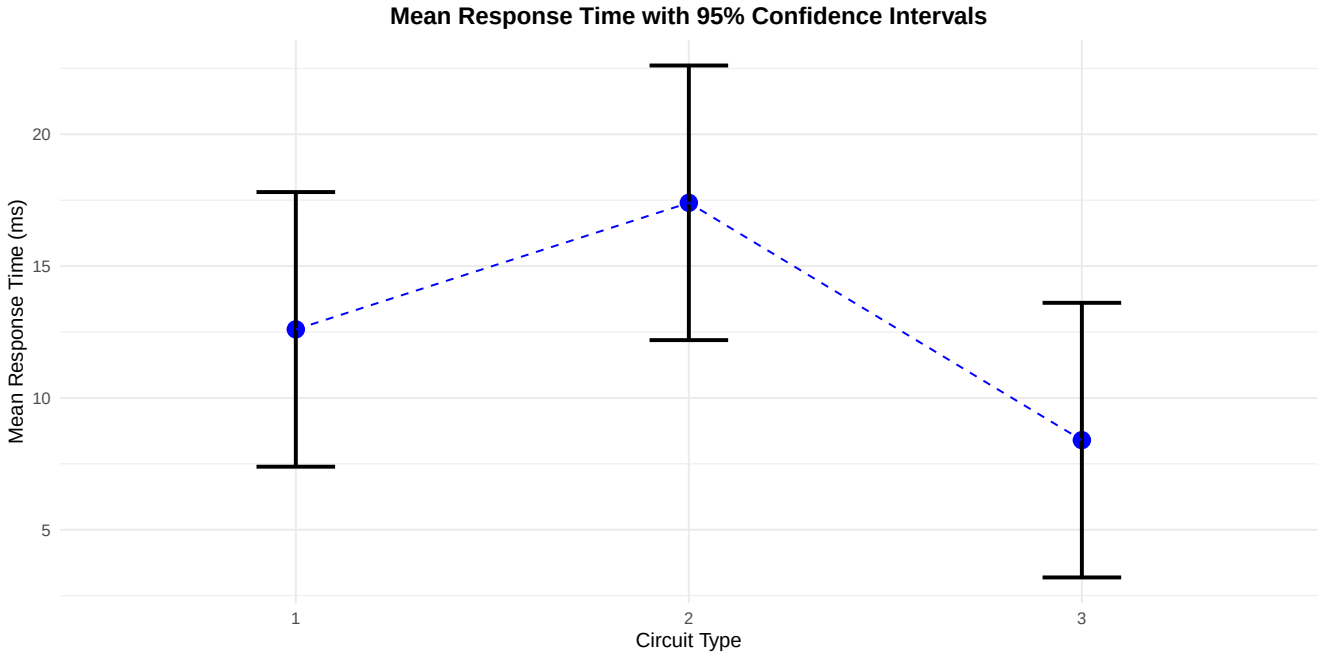
Equivalently, the **means model** is: $Y_{ij} = \mu_i + \epsilon_{ij}$

where $\mu_i = \mu + \tau_i$ is the mean response time for circuit type i.

Graphical Comparison







Conclusions from Graphical Analysis

1. The treatment means for response time differ across the three circuit types, with **Circuit Type 2** exhibiting the highest mean (approximately 17.5 ms), **Circuit Type 1** showing a moderate mean (approximately 12.5 ms), and **Circuit Type 3** demonstrating the lowest mean (approximately 8.5 ms).
2. The 95% confidence interval plot reveals that the intervals for Circuit Type 2 and Circuit Type 3 do not overlap, indicating a statistically significant difference between these two groups, whereas Circuit Type 1 overlaps with both, suggesting its differences may not be statistically significant.
3. The boxplot and strip plot highlight that Circuit Type 2 has the highest variability (widest spread), while Circuit Type 3 is the most consistent despite the presence of a single high outlier. Homogeneity of variance seems reasonable

4. **Strong visual evidence** that not all circuit types have the same mean response time

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Part (b)

Hypotheses:

Effects Model Form:

$$H_0 : \tau_1 = \tau_2 = \tau_3 = 0$$

$$H_a : \text{At least one } \tau_i \neq 0$$

Means Model Form (Equivalent):

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

$$H_a : \text{Not all means are equal}$$

One-Way ANOVA Table

Table 2: One-Way ANOVA for Circuit Response Time

Source	DF	SS	MS	F_value	P_value
Treatment (Circuit)	2	202.8	101.4	5.3936	0.0213
Error	12	225.6	18.8	NA	NA
Total	14	428.4	NA	NA	NA

```
## Critical value F(2, 12, 0.95): 3.8853
```

Decision Rule: Reject H_0 if $F > F_{critical}$ or if p-value $< \alpha$ (where $\alpha = 0.05$).

Conclusion: Since $F = 5.3936 > 3.8853$ and p-value $= 0.021327 < 0.05$, we **REJECT** H_0 at the 0.05 significance level. There is sufficient evidence to conclude that the three circuit types do **not** have the same mean response time.

Comparison with Part (a)

The hypothesis test **confirms the conclusions from the graphical analysis**:

- **Visual observation**: The plots showed clear differences between circuit types, especially Circuit 3 vs. the others
- **Statistical confirmation**: The ANOVA F-test provides strong statistical evidence ($p < 0.05$) that these differences are significant
- **Consistency**: Both approaches lead to the same conclusion that the circuit types have different mean response times
- **Reliability**: The combination of graphical and formal statistical analysis provides robust evidence for treatment effects

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Part (c)

For the effects model $Y_{ij} = \mu + \tau_i + \epsilon_{ij}$, the **Best Linear Unbiased Estimator (BLUE)** of the treatment mean $\mu_i = \mu + \tau_i$ is:

$$\hat{\mu}_i = \bar{Y}_{i.} = \frac{\sum_{j=1}^{n_i} Y_{ij}}{n_i}$$

This has expected value $E(\hat{\mu}_i) = \mu_i$ and variance $\text{Var}(\hat{\mu}_i) = \frac{\sigma^2}{n_i}$

```
## === CIRCUIT TYPE 3 ANALYSIS ===
##
## BLUE of mu_3 = ybar_3. = 8.4000 ms
##
## Standard Error: SE(ybar_3.) = sqrt(MSE/n_3) = sqrt(18.8000/5) = 1.9391
##
## Critical value: t(12, 0.975) = 2.1788
##
## 95% Confidence Interval for mu_3:
##   ybar_3. +/- t * SE(ybar_3.)
##   8.4000 +/- 2.1788 x 1.9391
##   (4.1751, 12.6249) ms
```

We are 95% confident that the true mean response time for Circuit Type 3 is between 4.18 and 12.62 milliseconds.

Part (d)

For the treatment contrast $\tau_i - \tau_j$, note that:

$$\tau_i - \tau_j = (\mu + \tau_i) - (\mu + \tau_j) = \mu_i - \mu_j$$

The **BLUE** is:

$$\widehat{\tau_i - \tau_j} = \bar{Y}_{i.} - \bar{Y}_{j.}$$

with variance:

$$\text{Var}(\bar{Y}_{i.} - \bar{Y}_{j.}) = \sigma^2 \left(\frac{1}{n_i} + \frac{1}{n_j} \right)$$

Calculation for $\tau_1 - \tau_2$

```
## === ANALYSIS OF tau_1 - tau_2 ===
##
## Overall mean: ybar.. = 12.8000
##
## Treatment effects (sum-to-zero constraint):
##   tau_hat_1 = ybar_1. - ybar.. = 12.6000 - 12.8000 = -0.2000
##   tau_hat_2 = ybar_2. - ybar.. = 17.4000 - 12.8000 = 4.6000
##   tau_hat_3 = ybar_3. - ybar.. = 8.4000 - 12.8000 = -4.4000
##
## BLUE of (tau_1 - tau_2):
##   tau_hat_1 - tau_hat_2 = -0.2000 - 4.6000 = -4.8000
```

```
##   Equivalently: ybar_1. - ybar_2. = 12.6000 - 17.4000 = -4.8000
##
## Standard Error:
##   SE = sqrt[MSE(1/n_1 + 1/n_2)] = sqrt[18.8000(1/5 + 1/5)] = 2.7423
##
## 95% Confidence Interval for (tau_1 - tau_2):
##   (ybar_1. - ybar_2.) +/- t * SE
##   -4.8000 +/- 2.1788 x 2.7423
##   (-10.7749, 1.1749)
```

We are 95% confident that the difference in treatment effects between Circuit 1 and Circuit 2 is between -10.77 and 1.17 ms. Since the interval CONTAINS 0, we cannot conclude that Circuit 1 and Circuit 2 have significantly different mean response times.

Problem 4

For the one-way ANOVA **effects model**:

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}, \quad i = 1, 2, \ldots, 5, \quad j = 1, 2, \ldots, n_i$$

A parametric function $\sum c_i \tau_i$ is **estimable** if and only if $\sum c_i = 0$ (under sum-to-zero constraint).

Summary Table

Below is the summary of the estimability analysis for the parameters.

Parameter	Estimable	BLUE	Variance
$\frac{\tau_1+\tau_2+\tau_3}{3} - \frac{\tau_4+\tau_5}{2}$	**Yes**	$\frac{\bar{y}_{1.}+\bar{y}_{2.}+\bar{y}_{3.}}{3} - \frac{\bar{y}_{4.}+\bar{y}_{5.}}{2}$	$\sigma^2 \left[\frac{1}{9n_1} + \frac{1}{9n_2} + \frac{1}{9n_3} + \frac{1}{4n_4} + \frac{1}{4n_5}\right]$
$\mu + \tau_2 + \tau_3 - \tau_4$	**Yes**	$\bar{y}_{2.} + \bar{y}_{3.} - \bar{y}_{4.}$	$\sigma^2 \left[\frac{1}{n_2} + \frac{1}{n_3} + \frac{1}{n_4}\right]$
$\tau_1 - \tau_5$	**Yes**	$\bar{y}_{1.} - \bar{y}_{5.}$	$\sigma^2 \left[\frac{1}{n_1} + \frac{1}{n_5}\right]$
$\tau_5 - \frac{\tau_1+\tau_2+\tau_3+\tau_4}{4}$	**Yes**	$\bar{y}_{5.} - \frac{\bar{y}_{1.}+\bar{y}_{2.}+\bar{y}_{3.}+\bar{y}_{4.}}{4}$	$\sigma^2 \left[\frac{1}{16n_1} + \frac{1}{16n_2} + \frac{1}{16n_3} + \frac{1}{16n_4} + \frac{1}{n_5}\right]$